Devisree Sumesh

Coimbatore, Tamil Nadu, India · devisreesumesh@gmail.com · +91-9995307906

Education

Amrita Vishwa Vidyapeetham *2023 – Present*
B.Tech, Electrical and Computer Science Engineering | Minor: Automation & Robotics | CGPA: ***8.59*** *Coimbatore, India*

Chavara Public School *2021 – 2023*
Class 12, CBSE | ***91.7%*** *Pala, India*

The Indian School *2009 – 2021*
Class 10, CBSE | ***97.4%*** *East Riffa, Bahrain*

Technical Skills

Languages	C, C++, Python, SQL
Robotics	ROS 2 (Jazzy), Gazebo, MoveIt2, URDF/Xacro, TF2, RViz2
Embedded	ESP32, STM32, PCA9685, ESP-NOW, LoRa (SX1278), IR2104, IRFZ44N, PIC16F877A
ML / DL	PyTorch, TensorFlow/Keras, Scikit-learn, Edge Impulse, YOLOv8, OpenCV, MediaPipe
Tools	STM32CubeIDE, MPLAB, LTSpice, MATLAB, Git, Google Colab
Coursework	Data Structures & Algorithms, AI & Edge Computing, IoT, Analog & Digital Electronics

Projects

Teleoperated FPV Robotic Arm with Surgical Vision Layer *2025 – Ongoing*

- Designing a 4-DOF wireless teleoperation system: potentiometer-based master arm (cardboard skeleton) → ESP32 $ESP-NOW$ slave ESP32 → PCA9685 servo controller driving MG996R and SG90 servos; ESP32-CAM mounted on wrist for live first-person surgical field view.
- Training YOLOv8n on CholecSeg8k surgical video dataset for real-time instrument detection; building Python + OpenCV inference pipeline for on-stream annotation of the wrist camera feed.

ROS 2 Jazzy — Robot Simulation & Motion Planning *2026*

- Built ROS 2 Jazzy development stack on Ubuntu 24.04: custom Python packages with `colcon` build system, publisher/subscriber nodes, service client–server, and action server–client communication patterns.
- Modelled robot in URDF, visualized TF2 transform trees in RViz2, simulated robot dynamics in Gazebo, and planned and executed joint-space trajectories via the MoveIt2 MoveGroup interface.

DC Motor PI Speed Control with Relay Auto-Tuning *2025*

- Implemented closed-loop DC motor speed control on ESP32: custom IRFZ44N MOSFET H-bridge with IR2104 bootstrap gate drivers (on-state drop <50 mV vs. ∼3 V in L298N), IR reflective sensor with `IRAM_ATTR` ISR for deterministic pulse counting.
- Implemented discrete PI controller (K_p=0.897, K_i=0.267) with forward-Euler integration and integral anti-windup (±500 counts); achieved steady-state speed of 84–96 RPM at 100 RPM reference, residual error bounded by 12 RPM measurement quantization floor.
- Implemented on-device Åström–Hägglund relay-feedback auto-tuning: software relay induces limit-cycle oscillation, measures T_u=3.17 s, computes Ziegler–Nichols gains automatically — eliminating need for offline plant identification.

Edge-Based Knee Rehabilitation Monitoring System *2025 – 2026*

- Acquired IMU time-series on ESP32; applied Edge Impulse DSP pipeline to train a compact neural network classifying 5 knee flexion–extension angle classes (20°–120°); deployed quantized model on-device.
- Built browser-based 3D knee joint visualizer (Three.js + Python serial-to-WebSocket bridge) and companion phone goniometer web app (JavaScript, DeviceOrientation API) for clinical ground-truth angle validation.

Portable Dual-Axis Solar Tracker *2025*

- Implemented the Solar Position Algorithm (SPA) in C on an STM32 ARM Cortex-M4 to compute sun angles using GPS/RTC input.
- Designed to map computed angles to SG90 servo PWM signal which positions the panel in direction of sunlight.
- Prepared a sample BOM as part of project documentation (IEEE format).

DriveWatch v3 — Real-Time Driver Fatigue Detection *2024*

- Built full-stack web app (Flask + SocketIO) for real-time driver monitoring: Eye Aspect Ratio (EAR) drowsiness detection via MediaPipe FaceMesh, phone distraction detection via COCO-SSD, composite fatigue scoring, and session-level sparkline trend overlay.

Intelligent Municipal Water Monitoring & Anomaly Detection *2025*

- Built multi-node IoT system: ESP32 streams flow, TDS, and ultrasonic level data over LoRa (SX1278) via custom register-level C++ driver (`OurLoRa.h`).

- Implemented cloud ML pipeline: 5-minute sliding-window feature extraction fed to Isolation Forest anomaly detector with EMA smoothing (α=0.3); full-stack dashboard (Node.js + Flask, Firebase RTDB, Chart.js dual-axis time-series).

Attention U-Net — Brain Tumour Segmentation *2025*

- Trained Attention U-Net on LGG MRI Dataset (110 patients, 3929 slices); achieved **Dice: 0.8388**, **IoU: 0.7270** on held-out test set.
- Implemented attention gate mechanism to learn soft spatial masks suppressing irrelevant encoder features before skip connections; trained with BCEDiceLoss and tuned binary prediction threshold on validation Dice and applied Grad-CAM on encoder layer activations to produce spatially interpretable heatmaps localizing tumour-discriminative features

Sleep Apnea Detection — 1D CNN on Respiratory Signals *2025*

- Built end-to-end apnea detection pipeline on PhysioNet APNEA-ECG dataset (35 recordings): 4th-order Butterworth bandpass (0.1–0.4 Hz), 30-second non-overlapping windows, per-window binary apnea/normal labels.
- Trained 1D CNN (3 conv layers, batch norm, global average pooling) in PyTorch; Leave-One-Patient-Out cross-validation across 35 subjects; per-patient normalization identified as critical preprocessing step after observing inter-patient baseline drift as primary failure mode.

Delhi-NCR Land Use Classification — Satellite ML *2025*

- Fine-tuned ResNet18 on Sentinel-2 multispectral satellite image patches labelled with ESA WorldCover classes; achieved **88.68% test accuracy** across 8 land cover categories.
- Built full geospatial ML pipeline: patch extraction, band normalization, stratified train/val/test split, per-class confusion matrix analysis; code at `DevisGit18/delhi-airshed-observation`.

μPAD Liver Enzyme Spectral Analysis *2025*

- Contributing to supervisor-led μPAD project (P.E. Resmi, Amrita): processing AS7341 11-channel spectral data from open paper strips and cassette-format strips for enzymatic colorimetric liver enzyme detection.
- Built 7-figure diagnostic plotting suite in Python (absorbance curves, channel heatmaps, blank-corrected signal comparison) to support calibration analysis; identified near-zero cassette absorbance as invalid blank reference artifact requiring protocol correction.

Awards

- **Sastra Pratibha** — Science India Forum (Vijnana Bharati), supported by Embassy of India, Bahrain & ISRO — 2015, 2018